SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 7

Case Study: Food Delivery Application – Negative Test Scenarios and Test Cases

AIM:

To identify and prepare negative test scenarios and test cases for the Food Delivery Application to verify that the system properly handles invalid registration details, incorrect login credentials, unavailable food items, invalid quantities, incorrect delivery addresses, failed payments, and unauthorized access while ensuring data accuracy, transaction security, and reliable order processing.

1. Problem Description

Negative testing checks invalid registration details, unavailable items, invalid quantities, incorrect addresses, failed payments and unauthorized access.

2. Functional Requirements

1. Users must register and log in using valid details.
2. Users must search for restaurants and food items.
3. Only available food items can be added to the cart.
4. Quantity and total price must be calculated correctly.
5. Valid delivery details are required.
6. Orders are confirmed only after successful payment.
7. Users can track eligible orders.

3. Negative Test Cases

Negative testing verifies that invalid, incomplete, or unexpected inputs are rejected safely with suitable messages.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Register without email	Leave email field blank	Required-field message displayed	Pass
NT02	Login with incorrect password	Enter correct email and wrong password	Login denied with error message	Pass
NT03	Search for nonexistent food	Enter a food name with no matching results	No matching items message displayed	Pass
NT04	Add unavailable food	Select an out-of-stock item	Item cannot be ordered	Pass

NT05	Enter zero quantity	Set quantity to 0	Quantity rejected	Pass
NT06	Enter quantity above stock	Request 10 items when only 5 are available	Quantity rejected with an appropriate message	Pass
NT07	Enter invalid PIN code	Enter letters or an invalid PIN	Invalid PIN rejected	Pass
NT08	Payment failure	Use a declined test payment	Order not confirmed; failure message displayed	Pass
NT09	Submit incomplete address	Leave mandatory address fields blank	Validation message displayed	Pass
NT10	Access order history without login	Open order history while logged out	User redirected to login	Pass

4. Detailed Execution of a Failed Test Case

Example: Select a test case that fails during actual execution. Record the precondition, steps, expected result, and actual result. Do not mark a test as failed unless the defect is observed.

5. Sample Bug Report

Field	**Details**
Bug ID	BR-001
Linked Test Case	Relevant failed negative test case
Severity / Priority	Assign according to impact and urgency
Expected Result	Invalid operation is rejected with a clear message
Actual Result	Application accepts or processes the invalid operation
Suggested Fix	Add server-side validation and repeat checks before final processing
Status	Open – awaiting developer investigation and correction

Conclusion

The negative test cases help verify that the Food Delivery Application meets its requirements and responds appropriately to the tested conditions.